

# Chapter 10: Understanding Correlation Output

## Interpreting Relationships Between Variables

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### 1 Overview

This reading prepares you for the lab session where you'll explore **correlations**. Correlation tells us whether two variables are related and how strongly. You'll learn to interpret correlation output and create meaningful scatterplots.

#### What You'll Learn

By the end of this lab, you'll be able to:

- Interpret correlation coefficients ( $r$  values)
- Understand what scatterplots reveal about relationships
- Read and interpret correlation test output
- Understand why correlation does not imply causation

#### Interactive Tool: Correlation Playground

Explore correlation interactively:

##### Launch Correlation Playground

Adjust correlation strength with sliders and see how scatterplots change in real-time. Build your intuition for what different  $r$  values look like.

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### 2 What Is Correlation?

**Correlation** measures the strength and direction of the linear relationship between two continuous variables.

- **Strength:** How tightly the data points cluster around a line (from weak to strong)
- **Direction:** Whether variables increase together (positive) or one increases as the other decreases (negative)

#### 2.1 The Correlation Coefficient ( $r$ )

The correlation coefficient  $r$  ranges from  $-1$  to  $+1$ :

| Value      | Interpretation                |
|------------|-------------------------------|
| $r = +1$   | Perfect positive relationship |
| $r = +0.7$ | Strong positive relationship  |
| $r = +0.3$ | Weak positive relationship    |
| $r = 0$    | No linear relationship        |
| $r = -0.3$ | Weak negative relationship    |
| $r = -0.7$ | Strong negative relationship  |
| $r = -1$   | Perfect negative relationship |

#### ! Direction vs Strength

The **sign** tells you direction; the **absolute value** tells you strength.  $r = -0.8$  is a stronger relationship than  $r = +0.3$ .

## 3 Reading Scatterplots

### 3.1 Why Visualise?

Before looking at numbers, **always** examine a scatterplot. It reveals:

- The direction of the relationship (positive, negative, or none)
- The strength of the relationship (tight cluster vs spread out)
- The shape (linear vs curved)
- Any unusual patterns or outliers

### 3.2 What to Look For

Each point represents one participant:

- **X-axis:** One variable
- **Y-axis:** The other variable

#### Pattern Guide:

| Pattern                   | Meaning               |
|---------------------------|-----------------------|
| Upward slope              | Positive relationship |
| Downward slope            | Negative relationship |
| No clear slope            | No relationship       |
| Tight cluster around line | Strong relationship   |
| Wide scatter              | Weak relationship     |

### 3.3 Visual Examples

Strong Positive ( $r \approx 0.8$ )



Weak Positive ( $r \approx 0.3$ )



No Relationship ( $r \approx 0$ )



## 4 Interpreting Correlation Output

When you run a correlation test, you'll see output like this:

```
Pearson's product-moment correlation

data:  reaction_time and time_estimate
t = 3.12, df = 48, p-value = 0.003
95 percent confidence interval:
 0.15  0.63
sample estimates:
      cor 
0.42
```

### 4.1 What Each Part Means

| Output                         | Meaning                                                |
|--------------------------------|--------------------------------------------------------|
| t = 3.12                       | The test statistic                                     |
| df = 48                        | Degrees of freedom (sample size - 2)                   |
| p-value = 0.003                | Probability of this r if there's truly no relationship |
| 95 percent confidence interval | Plausible range for the true correlation               |
| cor = 0.42                     | The sample correlation coefficient                     |

### 4.2 Key Questions to Answer

1. **What is r?** → 0.42 (medium positive correlation)
2. **Is it significant?** → Yes,  $p = 0.003 < 0.05$
3. **What's the CI?** → [0.15, 0.63] - the true r is likely in this range
4. **What does it mean?** → Higher reaction times are associated with higher time estimates

### 💡 Interpreting the Confidence Interval

A 95% CI of [0.15, 0.63] means we're fairly confident the true population correlation is somewhere in this range. Since it doesn't include 0, the correlation is statistically significant.

## 5 Correlation Strength Guidelines

Cohen (1988) suggested these benchmarks:

| $r$       |                 |  |
|-----------|-----------------|--|
| .10 - .29 | Small/weak      |  |
| .30 - .49 | Medium/moderate |  |
| .50+      | Large/strong    |  |

### ⚠️ Context Matters

These are rough guidelines. In some research areas,  $r = .30$  is typical and meaningful. In others,  $r = .50$  might be expected.

### 5.1 Understanding r-squared

Squaring  $r$  tells you the **proportion of variance explained**:

| $r$ | $r^2$ | Interpretation      |
|-----|-------|---------------------|
| .10 | .01   | 1% shared variance  |
| .30 | .09   | 9% shared variance  |
| .50 | .25   | 25% shared variance |
| .70 | .49   | 49% shared variance |

### ! Reality Check

A correlation of  $r = .42$  sounds reasonable. But  $r^2 = .18$  means one variable explains only 18% of the variance in the other. The remaining 82% is due to other factors!

## 6 Pearson vs Spearman

### 6.1 Pearson's $r$ (Most Common)

Use when:

- Both variables are continuous
- Relationship is approximately linear
- Data are roughly normally distributed

### 6.2 Spearman's $\rho$ ( $\rho$ )

Use when:

- Data are ordinal (ranked)
- Relationship is monotonic but not linear
- Data have outliers or are skewed

Both are interpreted the same way — they range from  $-1$  to  $+1$  with the same strength guidelines.

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## 7 Correlation Does NOT Mean Causation

### Critical Point

A correlation between X and Y does **NOT** mean X causes Y. This is possibly the most important lesson in research methods.

### 7.1 Why Not?

Three possible explanations for any correlation:

1. **X causes Y**: More sleep  $\rightarrow$  better performance
2. **Y causes X**: Better performance  $\rightarrow$  more sleep (less stress)
3. **Z causes both**: Good health  $\rightarrow$  more sleep AND better performance

Without experimental manipulation, we cannot determine which explanation is correct.

### 7.2 Classic Example

**Ice cream sales and drowning deaths are correlated.**

Does ice cream cause drowning? No! A third variable (hot weather) causes both:

- Hot weather  $\rightarrow$  more ice cream sales
- Hot weather  $\rightarrow$  more swimming  $\rightarrow$  more drowning

### 7.3 DataLab Example

Self-rating and throwing performance might be correlated.

Does thinking you're good make you perform better? Possible alternatives:

- Past experience → higher self-rating AND better performance
- Natural ability → higher self-rating AND better performance

We can't determine causation from correlation alone.

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## 8 Reporting Correlations (APA Style)

### 8.1 Template

Variable X and Variable Y were [significantly/not significantly] [positively/negatively] correlated,  $r(df) = .XX$ ,  $p = .XXX$ , 95% CI  $[.XX, .XX]$ .

### 8.2 Example

Reaction time and time estimation were significantly positively correlated,  $r(48) = .42$ ,  $p = .003$ , 95% CI  $[.15, .63]$ .

### 8.3 Elements to Include

- The variables being correlated
  - Direction (positive/negative)
  - Degrees of freedom in parentheses
  - The  $r$  value
  - The  $p$ -value
  - The 95% CI
- 

## 9 Preparation Checklist

Before arriving at the lab:

### Before You Arrive

- ☐ Read this chapter carefully
- ☐ Try the Correlation Playground app
- ☐ Review the lecture material on correlation
- ☐ Understand what  $r$  values mean (strength and direction)
- ☐ Be able to interpret example output

### 9.1 Questions to Consider

- Which DataLab variables might be correlated?
- What would a positive correlation between reaction time and time estimate mean?
- How would you explain correlation vs causation to a friend?

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## 10 Summary

| Concept                | Key Point                                              |
|------------------------|--------------------------------------------------------|
| <b>Correlation (r)</b> | Measures strength and direction of linear relationship |
| <b>Range</b>           | -1 (perfect negative) to +1 (perfect positive)         |
| <b>Scatterplot</b>     | Always visualise before interpreting numbers           |
| <b>r<sup>2</sup></b>   | Proportion of shared variance                          |
| <b>Significance</b>    | $p < .05$ suggests relationship exists in population   |
| <b>Causation</b>       | Correlation does NOT imply causation                   |

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## 11 Correlation Strength Quick Reference

| r   |              | Interpretation |     |
|-----|--------------|----------------|-----|
| .10 | Small        | .01            | 1%  |
| .20 | Small-medium | .04            | 4%  |
| .30 | Medium       | .09            | 9%  |
| .40 | Medium-large | .16            | 16% |
| .50 | Large        | .25            | 25% |
| .70 | Very large   | .49            | 49% |